Dynamic site-interconversion reduces the induction period of methanol conversion

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The conversion of methanol-to-olefins consists of several steps: (1) the induction period, (2) transition-regime, (3) steady-state, and (4) deactivation. Of these steps, the induction period determines the reaction efficiency. There are, however, scientific debates on the formation of the first C-C bond and primary olefins in the induction period. Up to 20 mechanisms have been proposed, including: (1) carbene, (2) carbon monoxide, (3) methane-formaldehyde, and (4) methoxy-methyl mechanisms. Determination of the mechanism of the formation of the first C-C bond is the first step towards controlling the induction period. After steady-state is achieved, it is necessary to obtain the mechanisms for the reduction of the induction period such that reaction efficiency is increased.

Methodology

Single and multiple-step response studies of dimethyl ether conversion over 10 mg of freshly activated ZSM-5 catalysts (Si/Al=25) at relatively low temperatures (300 °C and 330 °C for step-response), and at 1000 Pa in a temporal analysis of products (TAP) reactor were conducted. A particle-resolved transient microkinetic model was constructed by solving coupled non-linear partial differential equations that decouple the adsorption, desorption, mobility, and diffusion of species as well as convection, and dispersion. Also the model was used to investigate several mechanisms for the reduction of the induction period. These mechanisms include coverage, mass transfer, anomalous diffusion, fixed site-interconversion, and dynamic site-interconversion.

Results and discussion

A 44-min induction period is observed in the first step-response cycle. This induction period is reduced by 95% in the second and subsequent step response cycles. The methoxymethyl mechanism gives good agreement with experimental data

(figure 1) [1]. Of the five models investigated, the dynamic site interconversion model (figure 2) with surface methoxy species and adsorbed water allows for the reduction of the induction period over ZSM-5 catalysts [2]. To reduce the induction period, a steady flow of educt is required, as well as initial concentration of surface methoxy species, and adsorbed water. The latter can be achieved by methanol adsorption at different temperatures to generate the required surface coverages of surface methoxy species and adsorbed water. Consequently, a titration of the adsorbed species (surface methoxy species and adsorbed water) and a constant flow of educt (i.e., dimethyl ether) is required to reduce the original induction period by 95%, thereby leading to significant increases in reaction efficiency.

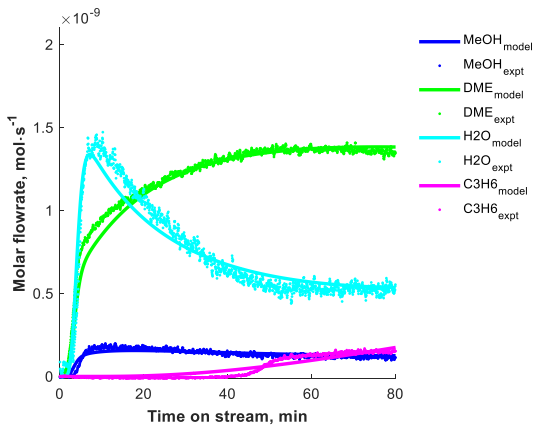

Figure 1: First step response cycle of the conversion of dimethyl ether to olefins over ZSM-5 (25) catalysts at 300 °C [1]

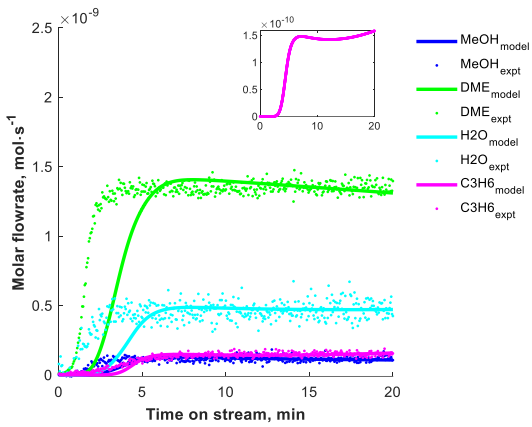

Figure 2: Second step response cycle of the conversion of dimethyl ether to olefins over ZSM-5 (25) catalysts at 300 °C [2]

Conclusions and significance

The methoxymethyl mechanism allows for the prediction of the induction period in the first step-response cycle. The induction period is reduced in the second and subsequent step response cycles through a combination of the methoxy methyl mechanism and the dynamic site-interconversion model between three site ensembles. The results of this work can be used to reduce the induction period, increase initial reaction efficiency, and improve catalytic activity.

References

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2. Omojola, T. ChemRXiv, 2024.DOI: https://doi.org/10.26434/chemrxiv-2024-wpr4m-v2.